9. CHIKUNGUNYA

Overview

Chikungunya is a re-emerging viral disease caused by **Chikungunya virus (CHIKV)**. It is primarily transmitted to humans through the bite of infected **Aedes mosquitoes**. The disease is usually self-limiting and is rarely fatal. Its most characteristic clinical features are **sudden fever, severe joint pain and rash**.

The disease can cause significant joint symptoms, and although mild joint manifestations generally resolve within a few weeks, more severe joint symptoms may persist for several months.

Key Facts

- **Causative agent:** Chikungunya virus (CHIKV).
- **Virus type:** Positive-sense, single-stranded RNA virus.
- **Family:** Togaviridae.
- **Genus:** Alphavirus.
- **Main vector:** Infected *Aedes* mosquitoes.
- Chikungunya is primarily transmitted through mosquito bites.
- **Incubation period:** Usually short; symptoms may begin approximately 48 hours after the bite and may take up to 12 days.
- The disease is generally **self-limiting** and rarely fatal.
- Severe joint pain is one of the most characteristic features.
- Chikungunya can occur without obvious symptoms, particularly in some children.

Transmission

The main transmission cycle is:

Infected person → Aedes mosquito → mosquito becomes infected → bite of another person → Chikungunya infection

- Chikungunya virus is most commonly transmitted through the bite of an infected *Aedes* mosquito.
- **Vertical transmission** is also possible.
- Viral levels in the blood are typically highest during the first two days of illness and decline thereafter.

- Because *Aedes* mosquitoes can also transmit other arboviral infections, clinical symptoms may overlap with diseases such as dengue.

Clinical Features and Symptoms

Chikungunya should be suspected when the characteristic combination of **fever, rash and joint pain** occurs.

Common clinical features include:

- Sudden onset of **fever**
- **Severe joint pain (arthralgia)**
- Multiple painful joints, often affecting both sides of the body
- Joint pain involving the **hands, wrists, ankles and feet**
- Joint stiffness
- Joint swelling in some patients
- Muscle pain
- Back pain
- Shoulder discomfort
- **Skin rash**
- Chills, particularly with high fever
- General constitutional symptoms

The rash may initially appear as flushing over the face and trunk and may develop into an erythematous skin eruption. It can involve the trunk, limbs, face, palms and soles.

Joint Symptoms

Joint involvement is a major feature of chikungunya.

- Joint pain may be **polyarticular and migratory**.
- Small joints of the hands, wrists, ankles and feet are commonly affected.
- Larger joints may also be involved.
- Severe pain can make walking difficult.
- Mild joint manifestations generally resolve within a few weeks.
- More severe joint symptoms may persist for **months**.

Rash and Skin Manifestations

Skin manifestations may begin with **flushing of the face and trunk** and progress to an erythematous rash.

The rash may occur on:

- Trunk
- Arms and legs
- Face
- Palms
- Soles

The rash generally fades or may undergo desquamation as the illness resolves.

Complications

Chikungunya is generally self-limiting, but complications can occasionally occur.

- **Meningoencephalitis** can occur infrequently.
- Neurological complications are particularly important in **newborns and people with pre-existing medical conditions**.
- Severe or prolonged joint manifestations may significantly affect mobility and quality of life.
- Deaths are rare.

Diagnosis

Chikungunya should be suspected clinically when there is the characteristic combination of:

Fever + Rash + Joint/Rheumatic Symptoms

Laboratory confirmation may be performed using:

- **Chikungunya-specific IgM antibody testing**
- **Virus isolation**
- **PCR-based detection**, particularly during the early stage of infection
- Molecular testing such as PCR can help identify active infection by detecting viral genetic material.

NCDC's Centre for Arboviral and Zoonotic Diseases lists the following Chikungunya diagnostic services:

- IgM ELISA

- IgM rapid diagnostic test
- IgG rapid diagnostic test
- PCR/qPCR for Chikungunya.

Treatment and Management

Chikungunya is generally **self-limiting**, and NCDC's material states that there is **no specific treatment** for the infection.

Management therefore focuses on supportive care:

- Adequate **rest**
- Adequate hydration and supportive care
- Management of fever and joint symptoms
- Avoid excessive physical activity during severe joint symptoms
- Gentle movement and mild exercise may help improve stiffness and morning joint pain.
- Persistent or severe joint symptoms require medical evaluation.

Treatment should be guided by a healthcare professional, particularly when symptoms are severe, prolonged or atypical.

Prevention and Control

Because chikungunya is mosquito-borne, prevention focuses on **avoiding mosquito bites and controlling mosquito breeding**.

Personal protection:

- Use mosquito repellents.
- Wear clothing that minimizes exposed skin.
- Use mosquito nets, particularly when resting during the daytime.
- Protect the home from mosquito entry where possible.

Mosquito control:

- Eliminate stagnant water.
- Remove containers and other places where mosquitoes can breed.
- Practise source reduction around homes and communities.
- Maintain environmental cleanliness.

- Support appropriate vector-control measures during outbreaks.

People suffering from chikungunya should also avoid mosquito exposure, because mosquitoes can acquire the virus from an infected person and subsequently contribute to transmission.

When to Seek Medical Attention

A person with suspected chikungunya should seek medical evaluation if:

- Fever is high or persistent.
- Joint pain is severe or progressively worsening.
- Symptoms continue for a prolonged period.
- There are neurological symptoms such as altered consciousness or seizures.
- The patient is a newborn or has significant underlying medical conditions.
- Symptoms are atypical or severe.
- There is concern for another illness such as dengue or another febrile infection.

Chikungunya: Quick Summary

Cause: Chikungunya virus
Spread: Infected *Aedes* mosquito bites
Incubation: Approximately 2–12 days
Main symptoms: Fever, severe joint pain, rash
Commonly affected joints: Hands, wrists, ankles and feet
Diagnosis: Clinical assessment + laboratory testing such as IgM and PCR
Treatment: Supportive care; no specific antiviral treatment
Prevention: Mosquito-bite protection + elimination of breeding sites
Prognosis: Usually self-limiting; joint symptoms may persist for months in some patients

Source: National Centre for Disease Control (NCDC), Ministry of Health & Family Welfare, Government of India. *Manual of Zoonotic Diseases of Public Health Importance (ZDPHI Manual 2025), Chapter 3: Arboviral Diseases – Chikungunya*